

AGF

The Pre-Execution Compliance Runtime for Regulated AI on AWS

Deterministic enforcement.

Sub-200ms ZK proofs.

Sovereign-grade assurance.

Five Strategic Outcomes for AWS

1



Remove the primary migration blocker

Convert stalled regulated workload deals into closed contracts by eliminating the cryptographic compliance gap.

\$2.4T banking IT spend — ~70% still on-prem

2



Convert compliance to deterministic enforcement

Replace probabilistic AI monitoring with ZK-proof pre-execution enforcement — the only architecture insurable at scale.

Uninsurable at scale vs. ZK-certifiable

3



Monetise verification at infrastructure layer

Proof-as-a-service revenue stream: per-proof fees, enterprise subscriptions, regulatory reporting premium tier.

Per-proof + subscription + premium tiers

4



Shorten 18–24 month regulated deal cycles

AGF provides certifiable assurance that removes board-level risk objections — the sole reason deals stall.

Boards cite compliance risk, not cost

5



Establish AWS as the sovereign AI runtime

Own the compliance layer beneath Bedrock and agentic AI — positioning AWS as the only cloud that can provably host regulated AI workloads.

Bedrock · GovCloud · Nitro native integration

Regulatory Inevitability — The Window Is Now



Performance Inflection

300×

proof speed improvement

GPU-native ZK proofs (SP1) now achieve sub-200ms generation — matching payment network latency. Previously impractical at 20–60s.

SP1 benchmarks, Succinct Labs 2025



Regulatory Binary

10%

global revenue penalty

EU AI Act full enforcement: Aug 2026. UK Online Safety Act: Jan 2026. Penalties are existential — boards now demand certified proof, not audit logs.

European Commission · Bank of England 2025



Agentic AI at Scale

Bn+

decisions/day requiring enforcement

Autonomous AI agents operating at machine speed require deterministic pre-execution enforcement. Post-hoc monitoring cannot meet binary compliance obligations.

McKinsey Global AI Report 2025

Three Walls Broken — Proof of Performance

172ms

Median proof latency

<240ms

99th percentile latency

\$0.0004

GPU cost per proof

~18 KB

Average proof size

01



Wall Broken: Performance

ZK proof generation previously took 20–60s — incompatible with real-time commerce. AGF leverages SP1 GPU-accelerated precompiles to achieve sub-200ms proof generation, matching Visa/Mastercard payment rail latency requirements.

Verification: External cryptographic audit in progress (Q2 2026). Patent filed PCT/GB2025/000XXX.

02



Wall Broken: Semantic Gap

Legacy AI produces probabilistic safety signals; statute law requires deterministic enforcement. AGF's Legal-Code Compiler (LCC) translates legal frameworks (EU AI Act, DORA, MiCA, UK FCA codes) directly into RISC-V bytecode and ZK circuits.

Result: Enforcement, not monitoring.

03



Wall Broken: Sovereign Vacuum

Fines previously treated as operational cost. New binary regimes impose 10% global revenue penalties and national market blocking — creating existential risk.

Boards now demand certified, mathematically provable compliance that can be presented to regulators as primary evidence.

Why This Architecture Cannot Be Replicated

Market Landscape

Category	Vendor	Limitation
Data Platforms	Snowflake / Databricks	Post-hoc analytics only
AI Governance	Palantir / IBM OpenPages	Monitoring, not enforcement
AI Model Guardrails	OpenAI / Anthropic	Probabilistic, prompt-layer
Cloud Providers	Azure / GCP	No deterministic compliance runtime
AGF	—	✓ Deterministic ZK pre-execution enforcement

No vendor offers deterministic, pre-execution, cryptographic compliance enforcement.

Replication Barriers

24–36 months

estimated internal build time for a hyperscaler

30+ jurisdictions

of sovereign regulatory mapping required

3 disciplines

cryptography + regulatory translation + AI runtime

First-mover

in cryptographic compliance primitives

Why Banks Can't Move to Cloud — How AGF Fixes It

\$2.4T

Global banking IT spend

~70%

Core workloads still on-prem

18–24mo

Avg regulated deal cycle

\$72B+

Potential AWS revenue uplift (5yr, 3% migration ↑)

The Blocker: Risk, Not Cost

- Boards require mathematical certainty at point of execution — not probabilistic signals.
- LLMs monitoring LLMs are uninsurable at scale, exposing infinite liability.
- Regulators now demand primary evidence, not audit logs.
- Typical regulated migration deal: 18–24 months and often fails at compliance sign-off.

AGF: The Surgical Fix

- Pre-execution ZK enforcement: non-compliant actions are blocked before they execute.
- Every action generates a verifiable cryptographic proof — primary regulatory evidence.
- Legal frameworks compile directly into enforcement logic (EU AI Act, DORA, MiCA, FCA).
- Removes the single blocker causing 18–24 month deal cycles and failures at sign-off.

What AGF Delivers — AWS-Native Architecture

AI Agent / Bedrock Workload



AGF Runtime · Compile → Enforce → Prove



AWS Infra · Nitro / GovCloud / Bedrock



Auditor / Regulator / Board Dashboard



Pre-Execution Enforcement

Rules compiled into ZK circuits. Non-compliant actions are blocked before execution — not flagged after. Deterministic by design.



Proof-of-Compliance

Every AI action produces a cryptographic ZK proof — mathematically verifiable by auditors, regulators, and boards. Primary regulatory evidence.



Jurisdiction-Aware Rule Compilation

EU AI Act, DORA, MiCA, UK FCA codes compile directly into enforcement logic via the Legal-Code Compiler (LCC). 30+ jurisdictions supported.

Six High-Value Use Cases Across Regulated Sectors



BANKING & REGTECH

ZK-AML

Zero-knowledge anti-money laundering enforcement. Transaction screening with cryptographic proof of compliance — no data exposure to counterparties. Estimated \$4.5B addressable in BFSI alone.



GOVERNMENT & SOVEREIGN

Sovereign AI Policy Enforcement

Air-gapped enforcement for ministerial AI decisions. Sovereign proof anchors provide tamper-proof audit trails for national regulators. Positions AWS GovCloud as default sovereign AI infrastructure.



HEALTHCARE & LIFE SCIENCES

Consent-Gated Clinical AI

Patient consent encoded as ZK circuits — AI systems are architecturally prevented from processing data beyond consent scope. HIPAA/GDPR primary evidence generated at every inference.



ENERGY & ESG

Carbon Credit Verification

Trustless carbon credit issuance with cryptographic provenance. Green finance taxonomy enforcement at the data layer — eliminates greenwashing liability for asset managers.



SUPPLY CHAIN & TRADE

Programmable Sanctions Enforcement

Trade transactions screened against OFAC/EU sanctions lists via ZK circuits — blocking non-compliant transfers pre-execution. Eliminates the enforcement gap in cross-border trade finance.



INSURANCE

Real-Time AI Risk Gating

Insurance underwriting AI constrained to regulatory risk bands pre-decision. Proof of actuarial compliance generated per policy — creating the first mathematically certifiable insurance product layer.

Build vs. Buy — The Strategic Clock Is Running

BUILD INTERNALLY

- ✗ **3–4 years**
to replicate cross-disciplinary stack
- ✗ **\$150–300M**
estimated internal development cost
- ✗ **30+ jurisdictions**
of regulatory mapping from scratch
- ✗ **Regulatory window**
missed — competitors lock in first
- ✗ **No first-mover**
advantage in cryptographic compliance primitives

VS

ACQUIRE AGF

- ✓ **Immediate**
category leadership in regulated AI compliance
- ✓ **Capture**
the 2026 regulatory enforcement window
- ✓ **Defensive moat**
vs. Microsoft Azure and Google Cloud
- ✓ **\$72B+ revenue**
uplift potential over 5 years (3% migration ↑)
- ✓ **AWS as**
the only cloud that can provably host regulated AI

Financial Analysis — DCF Scenarios & Unit Economics

Revenue Model

Tier	Model	Unit
Proof-as-a-Service	Per-proof fee	~\$0.0004/proof
Enterprise Base	Annual subscription	\$250K–2M/yr
Verification API	Usage-based	\$0.002/API call
Regulatory Reporting	Premium tier	\$500K+/yr

At 1B proofs/month → ~\$400M ARR potential

Assumptions: 10-yr DCF, 10% discount rate, 3% terminal growth

Acquisition Valuation Scenarios

CONSERVATIVE

\$2–5B

G7 public sector & BFSI only. Slow regulatory adoption. 5% take-rate on compliance savings.

Assumes limited AWS partnership, no mandate-level adoption.

BASE CASE

\$8–12B

Standard regulatory adoption + AWS partnership. Scale into global health & insurance.

2026 EU/UK enforcement drives BFSI migration; 3 major bank clients by 2027.

BULL CASE

\$20–25B

AGF becomes the mandated sovereign API for agentic AI. 1B+ monthly proofs. Universal logic layer.

Regulatory mandates in 3+ G7 jurisdictions; AWS exclusive distribution.

Key Risks & Mitigations — Pre-empting Due Diligence



ZK Compute Scaling

Medium Risk

GPU optimisation roadmap with SP1 precompiles. Benchmarked at 172ms median — on-target for payment rails. AWS Inferentia/Trainium partnership path.



Regulatory Fragmentation

Medium Risk

Jurisdiction-modular compiler architecture. Each legal framework is an independent module — new jurisdictions additive, not architectural changes. 30+ currently mapped.



Hyperscaler Internal Build

Low Risk

24–36 month head start. Cross-disciplinary stack (cryptography + regulatory translation + AI runtime) cannot be assembled rapidly. First-mover in cryptographic compliance primitives.



Proof Latency Spikes (99th pct)

Low Risk

Deterministic batching for non-real-time decisions. Real-time path benchmarked at <240ms (99th percentile). Graceful degradation to async proof for non-blocking decisions.



Regulatory Standard Uncertainty

Medium Risk

Modular rule compiler means standards updates are configuration changes, not re-architecture. Active engagement with EU AI Office, FCA, and Bank of England working groups.



Integration Complexity with AWS

Low Risk

Architecture built natively for cloud-native enforcement hooks. AWS Nitro enclaves, Bedrock API layer, and GovCloud alignment validated in technical design. Integration estimated at 6 months.

RECOMMENDATION

Pursue Exclusive Acquisition — Now

Regulatory momentum, GPU proof performance, and agentic AI adoption create a narrow, time-limited window. The compliance layer the cloud is missing is available today.

Remove the sole blocker preventing \$2.4T in banking IT from moving to AWS

Embed a category-defining ZK compliance runtime beneath Bedrock

Lock out Azure and GCP from the regulated AI workload segment

Monetise proof-as-a-service at every AI inference across the AWS estate

Valuation: \$8–12B base case · \$2–5B conservative · \$20–25B bull · 10-yr DCF, 10% discount rate

CONFIDENTIAL — Prepared for AWS Corporate Development, February 2026